2c. Content of chapters P1 to P8

Chapter P1: Radiation and waves

Overview

There are two key science ideas in this chapter – the first considers the uses of electromagnetic radiation and the possible health risks of radiation; both in nature and from technological devices, which are becoming of increasing concern. The second part of the topic considers a wave model for light and sound.

Topic P1.1 describes the model of radiation, an important scientific model for explaining how one object can affect another at a distance, and links this to the idea that all parts of the electromagnetic spectrum behave in this way. It then goes on to use the radiation model to explain how electromagnetic radiation behaves and to consider the risks and benefits of the technologies that use electromagnetic radiation. In some cases, misunderstanding the term 'radiation' generates unnecessary alarm. Through considering the evidence concerning the possible harmful effects of low-intensity microwave radiation from devices such as mobile phones, learners learn to evaluate reported health studies and interpret levels of risk.

Topic P1.2 introduces the idea that all bodies emit radiation to explain the greenhouse effect. Evidence for global warming is explored; scientific explanations for climate change draw on ideas about the way that radiation is emitted and absorbed by different materials. There is an opportunity to use both physical analogies and computer modelling to demonstrate the explanatory power of models in science.

All waves have properties in common and a wave model can be used to explain a great many phenomena, both natural and artificial. In Topic P1.3 the reflection and refraction of waves on water provide evidence that light and sound can be modelled as waves. Finally Topic P1.4 considers the behaviour of light and sound as they pass through a material interface, including refraction of light in lenses and prisms and the use of sound and ultrasound in imaging and detection.

Learning about radiation and waves before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- have observed waves on water, spring, and strings
- know the meaning of the terms longitudinal, transverse, superposition, and frequency, in the context of waves
- know that sound waves are longitudinal and need a medium to travel through and that sound travels at different speeds in solids, in water, and in air
- know that sound is produced when objects vibrate and that sound waves are detected by the vibrations they cause
- know some of the similarities and differences between light waves and waves in matter
- be able to use a ray model of light to describe and explain reflection in mirrors, refraction and dispersion by glass and the action of convex lenses
- know that light incident on a surface may be absorbed, scattered, or reflected, and that light transfers energy from a source to an absorber, where it may cause a chemical or electrical effect.

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Learning about Radiation and Waves at GCSE (9–1)

P1.1 What are the risks and benefits of using radiations?

<table>
<tr><th>Teaching and learning narrative</th><th colspan="2">Assessable learning outcomes
Learners will be required to:</th><th>Linked learning opportunities</th></tr>
<tr><td rowspan="4">A model of radiation can be used to describe and predict the effects of some processes in which one object affects another some distance away. One object (a source) emits radiation (of some kind). This spreads out from the source and transfers energy to other object(s) some distance away.

Light is one of a family of radiations, called the electromagnetic spectrum. All radiations in the electromagnetic spectrum travel at the same speed through space.</td><td>1.</td><td>describe the main groupings of the electromagnetic spectrum – radio, microwave, infrared, visible (red to violet), ultraviolet, X-rays and gamma rays, that these range from long to short wavelengths, from low to high frequencies, and from low to high energies</td><td rowspan="4">Practical work:
Estimate the speed of microwaves using a microwave oven.
Investigate how the intensity of radiation changes with distance from the source.</td></tr>
<tr><td>2.</td><td>recall that our eyes can only detect a very limited range of frequencies in the electromagnetic spectrum</td></tr>
<tr><td>3.</td><td>recall that all electromagnetic radiation is transmitted through space with the same very high (but finite) speed</td></tr>
<tr><td>4.</td><td>explain, with examples, that electromagnetic radiation transfers energy from source to absorber</td></tr>
<tr><td rowspan="2">When radiation strikes an object, some may be transmitted (pass through it), or be reflected, or be absorbed. When radiation is absorbed it ceases to exist as radiation; usually it heats the absorber.

Some types of electromagnetic radiation do not just cause heating when absorbed; X-rays, gamma rays and high energy ultraviolet radiation have enough energy to remove an electron from an atom or molecule (ionisation) which can then take part in other chemical reactions.</td><td>5.</td><td>recall that different substances may absorb, transmit, or reflect electromagnetic radiation in ways that depend on wavelength</td><td rowspan="2">Specification links:
Why are some materials radioactive? (P6.1)
How can radioactive materials be used safely (P6 .2).
How has our understanding of the atom developed over time? (C2.1)</td></tr>
<tr><td>6.</td><td>recall that in each atom its electrons are arranged at different distances from the nucleus, that such arrangements may change with absorption or emission of electromagnetic radiation, and that atoms can become ions by loss of outer electrons</td></tr>
</table>

P1.1 What are the risks and benefits of using radiations?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*
Exposure to large amounts of ionising radiation can cause damage to living cells; smaller amounts can causes changes to cells which may make them grow in an uncontrolled way, causing cancer. Oxygen is acted on by radiation to produce ozone in the upper atmosphere. This ozone absorbs ultraviolet radiation, and protects living organisms, especially animals, from its harmful effects. Radio waves are produced when there is an oscillating current in an electrical circuit. Radio waves are detected when the waves cause an oscillating current in a conductor. Different parts of the electromagnetic spectrum are used for different purposes due to differences in the ways they are reflected, absorbed, or transmitted by different materials. Developments in technology have made use of all parts of the electromagnetic spectrum; every development must be evaluated for the potential risks as well as the benefits (IaS4). Data and scientific explanations of mechanisms, rather than opinion, should be used to justify decisions about new technologies (IaS3).	7. recall that changes in molecules, atoms and nuclei can generate and absorb radiations over a wide frequency range, including: a) gamma rays are emitted from the nuclei of atoms b) X-rays, ultraviolet and visible light are generated when electrons in atoms lose energy c) high energy ultraviolet, gamma rays and X-rays have enough energy to cause ionisation when absorbed by some atoms d) ultraviolet is absorbed by oxygen to produce ozone, which also absorbs ultraviolet, protecting life on Earth e) infrared is emitted and absorbed by molecules
	8. describe how ultra-violet radiation, X-rays and gamma rays can have hazardous effects, notably on human bodily tissues
	9. give examples of some practical uses of electromagnetic radiation in the radio, microwave, infrared, visible, ultraviolet, X-ray and gamma ray regions of the spectrum
	10. recall that radio waves can be produced by, or can themselves induce, oscillations in electrical circuits

Linked learning opportunities
Ideas about Science: • Use the radiation model to predict and explain the behaviour of electromagnetic radiation (IaS3). **Practical work** • Investigate absorption, transmission and reflection of electromagnetic radiation e.g. absorption of ultraviolet by sunscreens, reflection and absorption of microwaves, or mobile phone signals. **Ideas about Science** • Discuss the different risks and benefits of technologies that use electromagnetic radiation (IaS4).

P1.2 What is climate change and what is the evidence for it?

Teaching and learning narrative	Assessable learning outcomes *Learners will be able to:*
All objects emit electromagnetic radiation with a principal frequency that increases with temperature. The Earth is surrounded by an atmosphere which allows some of the electromagnetic radiation emitted by the Sun to pass through; this radiation warms the Earth's surface when it is absorbed. The radiation emitted by the Earth, which has a lower principal frequency than that emitted by the Sun, is absorbed and re-emitted in all directions by some gases in the atmosphere; this keeps the Earth warmer than it would otherwise be and is called the greenhouse effect. One of the main greenhouse gases in the Earth's atmosphere is carbon dioxide, which is present in very small amounts; other greenhouse gases include methane, present in very small amounts, and water vapour. During the past two hundred years, the amount of carbon dioxide in the atmosphere has been steadily rising, largely the result of burning increased amounts of fossil fuels as an energy source and cutting down or burning forests to clear land. Computer climate models provide evidence that human activities are causing global warming. As more data is collected using a range of technologies, the model can be refined further and better predictions made (IaS3).	1. explain that all bodies emit radiation, and that the intensity and wavelength distribution of any emission depends on their temperatures **2. explain how the temperature of a body is related to the balance between incoming radiation, absorbed radiation and radiation emitted; illustrate this balance, using everyday examples including examples of factors which determine the temperature of the Earth**

Linked learning opportunities

Specification Links

- What is the evidence for climate change? (C1.2)

Practical work:

- Investigate climate change models – both physical models and computer models.

Ideas about Science

- Use ideas about the way science explanations are developed when discussing climate change (IaS3).
- Use ideas about correlation and cause when discussing evidence for climate change (IaS3).

P1.3 How do waves behave?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*
A wave is a regular disturbance that transfers energy in the direction that the wave travels, without transferring matter. For some waves (such as waves along a rope), the disturbance of the medium as the wave passes is at right-angles to its direction of motion. This is called a transverse wave. For other waves (such as a series of compression pulses on a slinky spring), the disturbance of the medium as the wave passes is parallel to its direction of motion. This is called a longitudinal wave. The speed of a wave depends on the medium it is travelling through. Its frequency is the number of waves each second that are made by the source. The wavelength of waves is the distance between the same points on two adjacent disturbances. The ways in which light and sound waves reflect and refract when they meet at an interface between two materials can be modelled with water waves. A wave model for light and sound can be used to describe and predict some behaviour of light and sound.	1. describe wave motion in terms of amplitude, wavelength, frequency and period 2. describe evidence that for both ripples on water surfaces and sound waves in air, it is the wave and not the water or air itself that travels 3. describe the difference between transverse and longitudinal waves 4. describe how waves on a rope are an example of transverse waves whilst sound waves in air are longitudinal waves 5. define wavelength and frequency 6. recall and apply the relationship between speed, frequency and wavelength to waves, including waves on water, sound waves and across the electromagnetic spectrum: wave speed (m/s) = frequency (Hz) × wavelength (m) M1a, M1c, M3c, M3d 7. a) describe how the speed of ripples on water surfaces and the speed of sound waves in air may be measured b) describe how to use a ripple tank to measure the speed/frequency and wavelength of a wave *PAG4* 8. a) describe the effects of reflection and refraction of waves at material interfaces b) describe how to measure the refraction of light through a prism *PAG8* c) describe how to investigate the reflection of light off a plane mirror *PAG8*

Linked learning opportunities

Ideas about Science

- Use the wave model to predict and explain the observed behaviour of light (IaS3).

Practical work:

- Carry out experiments to measure the speed of waves on water and the speed of sound in air.

P1.3 How do waves behave?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*
Refraction of light and sound can be explained by a change in speed of waves when they pass into a different medium; a change in the speed of a wave causes a change in wavelength since the frequency of the waves cannot change, and that this may cause a change in direction.	**9. recall that waves travel in different substances at different speeds and that these speeds may vary with wavelength**
	10. explain how refraction is related to differences in the speed of the waves in different substances
	11. recall that light is an electromagnetic wave
	12. recall that electromagnetic waves are transverse

Linked learning opportunities

P1.4 What happens when light and sound meet different materials? (*separate science only*)

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*		*Linked learning opportunities*
A beam of light is reflected from a smooth surface, such as a mirror, in a single beam which makes the same angle with the normal as the incident beam (specular reflection). Light is scattered in all directions from an uneven surface. Light is refracted at the boundary between glass (and water and Perspex) and air; this property is exploited in prisms and lenses. When a beam of white light is passed through a prism, the emerging light beam is spread out showing the colours of the spectrum. This can be explained using the wave model, different colours have different wavelengths; different wavelengths travel at different speeds when passing through glass, water or Perspex. What we perceive as white light is a mixture of different colours, ranging in wavelength from violet light (shortest visible wavelength) to red light (longest visible wavelength). A coloured filter works by allowing light of one or more wavelength through (transmission) and absorbing light of the other wavelengths. An object appears white if it scatters all the colours of light that fall on it, and black if it scatters none (and absorbs all). It appears coloured if it scatters light of some colours and absorbs light of other colours. Its observed colour is that of the light it scatters.	1.	construct and interpret two-dimensional ray diagrams to illustrate specular reflection by mirrors *qualitative only* M5a, M5b	**Practical work**: • Trace light rays through glass blocks, prisms and lenses and when reflected from mirrors. • Investigate the effects of looking at coloured object through coloured filters. • Investigate the transmission of light and sound across interfaces.
	2.	construct and interpret two-dimensional ray diagrams to illustrate refraction at a plane surface and dispersion by a prism *qualitative only* M5a, M5b	
	3.	use ray diagrams to illustrate the similarities and differences between convex and concave lenses *qualitative only*	
	4.	describe the effects of transmission, and absorption of waves at material interfaces	
	5.	explain how colour is related to differential absorption, transmission, and scattering	
	6.	**describe, with examples, processes in which sound waves are transmitted though solids**	
	7.	**explain that transmission of sound through the bones in the ear works over a limited frequency range, and the relevance of this to human hearing**	

P1.4 What happens when light and sound meet different materials? (*separate science only*)

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*	*Linked learning opportunities*
Sound travels better through solids and liquids than through air. The small bones in the middle ear transmit the sound waves from the air outside to the inner ear. The bones transmit frequencies most efficiently in the range 1 kHz and 3 kHz. **The ways in which sound waves are transmitted, reflected and refracted as they pass through liquids and solids are exploited in ultrasound imaging in medicine, in exploring the structure of the Earth and in using SONAR to explore under water.**	**8. explain, in qualitative terms, how the differences in velocity, absorption and reflection between different types of waves in solids and liquids can be used both for detection and for exploration of structures which are hidden from direct observation, notably:** **a) in our bodies (ultrasound imaging)** **b) in the Earth (earthquake waves)** **c) in deep water (SONAR)**	
	9. show how changes, in speed, frequency and wavelength, in transmission of sound waves from one medium to another, are inter-related M1c, M3c	